

Title & Project Overview

•**Title:** AECO Site Turnover Defect Detection using YOLO11

•**Subtitle:** Automated Visual Quality Assurance & Punchlist Inspection

•**Key Bullet Points:**

- **Objective:** Automate detection of 5 critical handover defects across construction site turnover images.
- **Business Value:** Replaces slow, error-prone manual punchlists with real-time AI-assisted screening.
- **Success Target:** High Recall (≥ 0.70) on major defects to minimize post-turnover warranty risks.

Dataset Composition & Splits

•**Title:** Dataset Overview & Class Distribution

•**Key Bullet Points:**

- **Source:** Hosted on [Roboflow Universe](#) under [CC BY 4.0](#).
- **Splits:**
 - **Train:** 28 images (80%)
 - **Validation:** 3 images (10%)
 - **Test:** 5 images (10%)
- **5 Target Classes:** cracked jamb, cracked tile, moisture stain, paint defect, uneven joint.

Model Architecture & Hyperparameters

•**Title:** Model Architecture & Training Setup

•**Key Bullet Points:**

- **Backbone:** YOLO11 Nano (yolo11n.pt) chosen for lightweight, edge-deployable real-time inference.
- **Training Parameters:** 50 Epochs, Batch Size = 16, Image Size = 640×640 .
- **Hardware Runtime:** Executed on Google Colab Tesla T4 GPU (~15 minutes execution time).

Metrics Summary & Validation Results

•**Title:** Performance Comparison: Baseline vs. Fine-Tuned

•**Metrics Table:**

Model Stage	Precision (P)	Recall (R)	mAP@50	mAP@50-95
Baseline (Zero-Shot YOLO11n)	0.0000	0.0000	0.0000	0.0000
Fine-Tuned YOLO11n (50 Epochs)	0.0080	1.0000	0.0684	0.0350

•**Key Takeaway:** Domain fine-tuning achieves 100% Recall on validation instances, but low Precision highlights a high false-positive rate on background shadows and surface textures.

Visual Evidence & Predictions

•**Title:** Prediction Gallery & Model Artifacts

•**Key Bullet Points:**

- **Validation Output:** Model successfully locates defects like moisture stain and uneven joint.
- **Repository Artifacts:** All model weights (best.pt), training curves, and validation prediction logs are stored under the /results/ folder on GitHub.

Failure Mode Analysis

•**Title:** Error Analysis: False Positives & Negatives

•**Key Bullet Points:**

- **False Positives:** Surface shadows, tile grout line depth variations, and wall dirt/scuffs misidentified as defects.
- **False Negatives:** Micro-cracks on door frames and faint water ring outlines blurred during 640×640 image downsampling.
- **Actionable Next Steps:** Apply CLAHE lighting augmentations, tile high-res crops, and balance class samples.

Governance & Compliance Checklist

•**Title:** AI Governance, Privacy & Licensing

•**Key Bullet Points:**

- **Privacy:** All site photos anonymized to exclude faces, worker IDs, and unit numbers.
- **Data Minimization:** Restricted exclusively to target defect surface areas.
- **Limitations:** Strictly an operational screening assistant; **not** certified for structural load-bearing approvals.
- **License:** Code under [MIT License](#); dataset under [CC BY 4.0](#).